Martingales — Exam (set 1)

Name: ______________________________ **Date:** ____________________

Duration: 3 hours. You may use any result from the course by citing it precisely. The questions can often be solved independently. Marks out of 100 (indicative). *Answer in the space provided; the rigour of your justifications matters.*

Problem I — Likelihood Ratio Martingale, Hellinger Affinity, and Change of Reference Law (18 pts)

(indicative time: 35 min)

We work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $(\mathsf{X}, \mathcal{A}, \nu)$ be a σ-finite measure space, and let

$$f, g : \mathsf{X} \longrightarrow [0, \infty[$$

be two probability densities with respect to ν, that is, $\int_{\mathsf{X}} f \, d\nu = \int_{\mathsf{X}} g \, d\nu = 1$. We assume that

$$f(x) > 0 \quad \text{for all } x \in \mathsf{X},$$

so that the quotient g/f is everywhere well defined.

Let $(X_n)_{n\geq 1}$ be a sequence of *independent and identically distributed* random variables, taking values in $(\mathsf{X}, \mathcal{A})$, whose common law admits the density f with respect to ν; in other words, for every measurable function $h : \mathsf{X} \to [0, \infty[$,

$$\mathbb{E}[h(X_1)] = \int_{\mathsf{X}} h(x) \, f(x) \, d\nu(x).$$

Set $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$ for $n \geq 1$. Define the *likelihood ratio* by $L_0 = 1$ and, for $n \geq 1$,

$$L_n \;=\; \prod_{i=1}^{n} \frac{g(X_i)}{f(X_i)}.$$

Finally set the *Hellinger affinity* constant

$$\rho \;=\; \int_{\mathsf{X}} \sqrt{f(x)\, g(x)} \; d\nu(x) \;\in [0, \infty[.$$

Question 1 (integrability and martingale).

(a) Show that for every $n \geq 0$, $L_n \geq 0$ and $L_n \in L^1(\mathbb{P})$ with $\mathbb{E}[L_n] = 1$.

(b) Show that $(L_n)_{n\geq 0}$ is a martingale with respect to the filtration $(\mathcal{F}_n)_{n\geq 0}$.

Question 2 (Hellinger affinity and supermartingale).

(a) Show that $0 \leq \rho \leq 1$ and that for every $n \geq 0$,

$$\mathbb{E}\Big[\sqrt{L_n}\Big] = \rho^n.$$

(b) Deduce, using the conditional Jensen inequality, that the process $\big(\sqrt{L_n}\big)_{n\geq 0}$ is a supermartingale. (Specify which convex / concave function is used.)

(c) Assume in addition that $\nu(\{x : f(x) \neq g(x)\}) > 0$. Show that then $\rho < 1$, and deduce that $L_n \to 0$ almost surely as $n \to \infty$. *(The almost sure convergence of nonnegative supermartingales may be used, and is admitted here.)*

Question 3 (change of reference law). Define a second probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ under which $(X_n)_{n\geq 1}$ is i.i.d. with density g with respect to ν. We assume here moreover that $g(x) > 0$

for all $x \in \mathsf{X}$, and we set

$$M_n = \frac{1}{L_n} = \prod_{i=1}^{n} \frac{f(X_i)}{g(X_i)}.$$

(a) Show that $(M_n)_{n\geq 0}$ is a martingale *under* $\mathbb{Q}$ with respect to $(\mathcal{F}_n)_{n\geq 0}$.

(b) Set $\chi = \int_{\mathsf{X}} \frac{g(x)^2}{f(x)}\, d\nu(x)$, and assume $\chi < \infty$. Show that $\chi \geq 1$ and that, *under* $\mathbb{Q}$,

$$\mathbb{E}_{\mathbb{Q}}[L_{n+1} \mid \mathcal{F}_n] = \chi\, L_n.$$

Deduce that $(L_n)_{n\geq 0}$ is a submartingale under $\mathbb{Q}$.

Question 4 (convexity). Let $\varphi : [0,\infty[\to \mathbb{R}$ be a convex function such that $\varphi(L_n) \in L^1(\mathbb{P})$ for every n. Show (under $\mathbb{P}$) that $\big(\varphi(L_n)\big)_{n\geq 0}$ is a submartingale, and that the sequence of expectations $n \mapsto \mathbb{E}[\varphi(L_n)]$ is nondecreasing. Deduce that, when $\mathbb{E}[L_n \log^+ L_n] < \infty$ for every n, the sequence $n \mapsto \mathbb{E}\big[L_n \log^+ L_n\big]$ is nondecreasing. (Justify the convexity of the function used.)

Problem II — Ruin Probability of a Skip-Free-Downward Random Walk with Positive Drift **(20 pts)**

(indicative time: 40 min)

Let $p \in]0,1[$ and $q = 1-p$. Let $(X_n)_{n\in\mathbb{N}^*}$ be a sequence of independent and identically distributed random variables with law

$$\mathbb{P}(X_n = +2) = p, \qquad \mathbb{P}(X_n = -1) = q.$$

Set $S_0 = 0$, $S_n = X_1 + \cdots + X_n$ for $n \geq 1$, and $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$ (with $\mathcal{F}_0 = \{\varnothing, \Omega\}$). The walk $(S_n)_{n\in\mathbb{N}}$ goes up by jumps of $+2$ but goes down only by jumps of -1: it is said to be *skip-free downward*. Throughout the problem we assume that the drift is strictly positive:

$$\mu := \mathbb{E}[X_1] = 2p - q = 3p - 1 > 0, \qquad \text{that is, } p > \tfrac{1}{3}.$$

For an integer $a \geq 1$, denote by

$$\tau_{-a} = \inf\{n \geq 0 : \ S_n = -a\} \qquad (\inf \varnothing = +\infty)$$

the first time the walk reaches the deficit level $-a$.

Question 1. Show that τ_{-a} is a stopping time with respect to $(\mathcal{F}_n)_{n\in\mathbb{N}}$. Verify moreover that, on the event $\{\tau_{-a} < \infty\}$, one has $S_{\tau_{-a}} = -a$ exactly (in other words, the walk cannot "jump over" the level $-a$).

Question 2. We seek $\theta \in \mathbb{R}$ such that the process

$$M_n := e^{\theta S_n} \qquad (n \in \mathbb{N})$$

is a martingale with respect to $(\mathcal{F}_n)_{n\in\mathbb{N}}$.

(a) Show that $(M_n)_{n\in\mathbb{N}}$ is a martingale if and only if $y := e^\theta$ satisfies $p\,y^3 - y + q = 0$.

(b) Show that this cubic equation admits a unique root $y_* \in]0,1[$, given by

$$y_* = \frac{-p + \sqrt{p^2 + 4pq}}{2p}.$$

From now on, set $\theta_* = \ln y_* < 0$ and $M_n = y_*^{S_n}$.

Question 3 (ruin probability). We want to compute the ruin probability $\rho := \mathbb{P}(\tau_{-a} < \infty)$.

(a) Justify that the stopped process $(M_{n\wedge\tau_{-a}})_{n\in\mathbb{N}}$ is a bounded martingale, and specify a deterministic upper bound $C < \infty$ such that $0 \le M_{n\wedge\tau_{-a}} \le C$ for all n.

(b) Show that $M_{n\wedge\tau_{-a}} \to y_*^{-a}\,\mathbf{1}_{\{\tau_{-a}<\infty\}}$ almost surely as $n \to \infty$. *(You may take for granted that, by the law of large numbers, $S_n \to +\infty$ a.s. on $\{\tau_{-a} = \infty\}$.)*

(c) Deduce, carefully justifying the passage to the limit, that

$$\boxed{\rho = \mathbb{P}(\tau_{-a} < \infty) = y_*^{a}}$$

Question 4 (exit time from a band). Now let $b \ge 1$ be an integer and

$$\tau = \inf\{n \ge 0 : \ S_n \le -a \ \text{ or } \ S_n \ge b\}.$$

(a) Show that there exist $q_0 \in]0,1[$ and $k \in \mathbb{N}^*$ such that, for all $n \in \mathbb{N}$,

$$\mathbb{P}(\tau \leq n + k \mid \mathcal{F}_n) \geq q_0 \quad \text{almost surely on } \{\tau > n\}.$$

Deduce that $\mathbb{E}[\tau] < \infty$.

(b) Justify that Wald's identity $\mathbb{E}[S_\tau] = \mu\,\mathbb{E}[\tau]$ applies to this stopping time, and deduce the expression of $\mathbb{E}[\tau]$ in terms of $\mathbb{E}[S_\tau]$ and p.

Question 5. We now take the numerical values $p = \frac{2}{5}$ and $a = 3$.

(a) Compute explicitly y_* and the ruin probability $\rho = \mathbb{P}(\tau_{-3} < \infty)$.

(b) A gambler with zero initial fortune follows the walk (S_n). He *wins* the game if the walk ever reaches the deficit -3; otherwise it runs off to $+\infty$ and he *loses*. Does the above computation show that $\mathbb{P}(\tau_{-3} < \infty) < 1$ even though the drift is positive? Comment briefly on the role of the asymmetry of the jumps ("skip-free downward") in the possibility of having $\rho = 1$ or $\rho < 1$.

Problem III — Random Products of Rademacher Factors: Kakutani's Dichotomy **(21 pts)**

(indicative time: 40 min)

Let $(\xi_n)_{n\in\mathbb{N}^*}$ be a sequence of independent and identically distributed random variables with Rademacher distribution:

$$\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = \tfrac{1}{2} \qquad (n \in \mathbb{N}^*).$$

We are given a deterministic sequence $(a_n)_{n\in\mathbb{N}^*}$ of reals with $a_n \in [0,1)$ for all n. Set $\mathcal{F}_0 = \{\emptyset, \Omega\}$, $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ for $n \geq 1$, and define the *random product*

$$M_0 = 1, \qquad M_n = \prod_{k=1}^{n} \bigl(1 + a_k\, \xi_k\bigr) \qquad (n \in \mathbb{N}^*).$$

1. Show that $(M_n)_{n\in\mathbb{N}}$ is a positive martingale with respect to $(\mathcal{F}_n)_{n\in\mathbb{N}}$, and that $\mathbb{E}[M_n] = 1$ for all n. Deduce that $M_n \to M_\infty$ almost surely, where M_∞ is a positive integrable random variable.

2. Compute $\mathbb{E}[M_n^2]$ in terms of the a_k. Deduce that:

(a) if $\sum_{k\geq 1} a_k^2 < \infty$, then $(M_n)_{n\in\mathbb{N}}$ is bounded in L^2, hence converges to M_∞ in L^2 (and in particular in L^1);

(b) in this case, compute $\mathbb{E}[M_\infty]$ and $\mathbb{E}[M_\infty^2]$.

3. Assume still that $\sum_{k\geq 1} a_k^2 < \infty$. Show that the martingale $(M_n)_{n\in\mathbb{N}}$ is uniformly integrable and that it is *closed*, that is, $M_n = \mathbb{E}[M_\infty \mid \mathcal{F}_n]$ for all $n \in \mathbb{N}$.

4. *(Maximal bound.)* Assume here that $a_k = a$ is constant for all k, with $a \in [0,1)$ fixed; thus $\sum_k a_k^2 = \infty$ as soon as $a > 0$. Using Doob's maximal inequality, show that for all $\lambda > 0$,

$$\mathbb{P}\Big(\max_{0\leq k\leq n} M_k \geq \lambda\Big) \leq \frac{1}{\lambda}, \qquad \text{then} \qquad \mathbb{P}\Big(\sup_{k\geq 0} M_k \geq \lambda\Big) \leq \frac{1}{\lambda}.$$

Deduce that $\sup_{k\geq 0} M_k < \infty$ almost surely.

5. *(The degenerate case.)* In this question assume $a_k = a$ constant with $a \in (0,1)$, so that $\sum_{k\geq 1} a_k^2 = \infty$.

(a) By considering $\log M_n = \sum_{k=1}^n \log(1 + a\,\xi_k)$, compute $\mathbb{E}[\log(1 + a\,\xi_k)]$. You may use that for $x \in [0,1)$, $\frac{1}{2}\log(1-x^2) \leq -\frac{x^2}{2}$. Using the strong law of large numbers, deduce that $M_\infty = 0$ almost surely.

(b) Conclude that, in this case, the martingale $(M_n)_{n\in\mathbb{N}}$ is *not* uniformly integrable, that it does not converge in L^1, and that it is not closed.

6. *(General case, optional.)* We return to the general case $a_k \in [0,1)$ with $\sum_{k\geq 1} a_k^2 = \infty$. Set

$$c_k = \mathbb{E}\big[\sqrt{1+a_k\xi_k}\big] = \tfrac{1}{2}\big(\sqrt{1+a_k} + \sqrt{1-a_k}\big), \qquad N_n = \prod_{k=1}^{n} \frac{\sqrt{1+a_k\xi_k}}{c_k}.$$

Verify that (N_n) is a positive martingale with expectation 1, that $1 - c_k^2 \geq a_k^2/4$, and deduce that $\prod_{k\geq 1} c_k = 0$, and then that $M_\infty = 0$ almost surely.

Problem IV — Geometric Bienaymé–Galton–Watson process: explicit law of the martingale limit W (21 pts)

(indicative time: 40 min)

We consider a Bienaymé–Galton–Watson process $(Z_n)_{n\geq 0}$ started from a single ancestor, $Z_0 = 1$. Each individual, independently of the others, gives birth to a random number of offspring with *geometric* distribution on $\{0, 1, 2, \dots\}$:

$$\mathbb{P}(\xi = k) = (1-p)\,p^k, \qquad k = 0, 1, 2, \dots,$$

where $p \in \left(\frac{1}{2}, 1\right)$ is a fixed parameter. We write

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_i^{(n)},$$

where the $\xi_i^{(n)}$ are i.i.d. with the same law as ξ and independent of the past, with the convention $\sum_{i=1}^{0} = 0$. Set $f(s) = \mathbb{E}[s^\xi]$ for $s \in [0,1]$, $m = \mathbb{E}[\xi]$, and $\mathcal{F}_n = \sigma(Z_0, \dots, Z_n)$. Recall that the generating function of Z_n (with $Z_0 = 1$) is the n-th compositional iterate $f_n := \mathbb{E}[s^{Z_n}] = f^{\circ n}$.

Question 1 (warm-up).

(a) Show that $f(s) = \dfrac{1-p}{1-ps}$ for $s \in [0,1]$, and compute $m = \mathbb{E}[\xi]$ as well as the variance $\sigma^2 = \mathrm{Var}(\xi)$. Check that $m > 1$.

(b) Recall why $W_n := Z_n/m^n$ is a positive $(\mathcal{F}_n)$-martingale, and deduce that it converges almost surely to a random variable $W \geq 0$ with $\mathbb{E}[W] \leq 1$.

Question 2 (extinction probability and closed form of f_n). Let $q = \mathbb{P}(\exists\, n : \; Z_n = 0)$ be the extinction probability.

(a) State the characterization of q as a fixed point of f, then solve explicitly the equation $f(s) = s$ and deduce that

$$q = \frac{1-p}{p} = \frac{1}{m}.$$

(b) Prove the conjugation identity

$$\frac{f_n(s)-1}{f_n(s)-q} = m^n\,\frac{s-1}{s-q}, \qquad s \in [0,1).$$

Deduce the closed form

$$f_n(s) = \frac{(m^n-1)-(m^n-m)\,s}{(m^{n+1}-1)-(m^{n+1}-m)\,s}.$$

Question 3 (explicit law of the limit W).

(a) Show that the Laplace transform of W equals

$$\mathbb{E}\Big[e^{-\lambda W}\Big] = \lim_{n\to\infty} f_n\big(e^{-\lambda/m^n}\big) = \frac{\lambda+m-1}{m\lambda+m-1}, \qquad \lambda \geq 0.$$

(b) Deduce the law of W: show that

$$\mathbb{P}(W=0) = \frac{1}{m} = q, \qquad \text{and that, conditionally on } \{W>0\},\ \ W \sim \mathcal{E}\Big(\tfrac{m-1}{m}\Big)$$

(exponential distribution with parameter $(m-1)/m$). Check in particular that $\mathbb{E}[W]=1$: the martingale W_n is therefore *non-degenerate.*

Question 4 (L^2 convergence and variance).

(a) Establish the recurrence relation $\mathbb{E}[Z_{n+1}^2 \mid \mathcal{F}_n] = m^2 Z_n^2 + \sigma^2 Z_n$, and deduce that

$$\mathbb{E}[W_{n+1}^2] = \mathbb{E}[W_n^2] + \frac{\sigma^2}{m^{n+2}}.$$

(b) Deduce that $\sup_n \mathbb{E}[W_n^2] < \infty$, that $W_n \to W$ in L^2, and compute $\mathrm{Var}(W)$. Check that the value obtained coincides with the one arising from the law of W found in Question 3.

Problem V — Brownian Motion with Negative Drift: Crossing Probability and the Law of the Maximum (20 pts)

(indicative time: 38 min)

Let $(B_t)_{t\in\mathbb{R}_+}$ be a standard Brownian motion in $\mathbb{R}$, starting from $B_0 = 0$, defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ equipped with its natural filtration $(\mathcal{F}_t)_{t\in\mathbb{R}_+}$, where $\mathcal{F}_t = \sigma(B_s\,;\, s \in [0,t])$. We fix two strictly positive reals $a > 0$ and $c > 0$, and we consider the *Brownian motion with negative drift*

$$X_t \;=\; B_t - c\,t, \qquad t \in \mathbb{R}_+.$$

We define the hitting time of the linear barrier $t \mapsto a + ct$ by the Brownian motion:

$$\tau_a \;=\; \inf\{\, t \in \mathbb{R}_+ \;:\; B_t = a + c\,t \,\} \;=\; \inf\{\, t \in \mathbb{R}_+ \;:\; X_t = a \,\},$$

with the convention $\inf \varnothing = +\infty$. Finally, we set

$$S \;=\; \sup_{t\in\mathbb{R}_+} X_t \;=\; \sup_{t\in\mathbb{R}_+} (B_t - c\,t).$$

For $\lambda \in \mathbb{R}$, recall the *exponential martingale*

$$M_t^\lambda \;=\; \exp\Big(\lambda B_t - \tfrac{\lambda^2}{2}\,t\Big), \qquad t \in \mathbb{R}_+.$$

Question 1 (preliminaries).

(a) Show that for every $\lambda \in \mathbb{R}$, the process $(M_t^\lambda)_{t\in\mathbb{R}_+}$ is a martingale with respect to $(\mathcal{F}_t)_{t\in\mathbb{R}_+}$, and that $\mathbb{E}[M_t^\lambda] = 1$ for all t.

(b) We admit (a corollary from the chapter on Brownian motion) that $B_t/t \to 0$ almost surely as $t \to +\infty$. Show that almost surely $\lim_{t\to+\infty} X_t = -\infty$. Deduce that the random variable S is finite almost surely and that $S \geq 0$.

Question 2 (crossing probability).

(a) Justify that τ_a is a stopping time, then show, by applying the optional stopping theorem (the *bounded stopping time* version $\tau_a \wedge t$) to the martingale (M_t^λ) with the choice $\lambda = 2c > 0$, that for all $t \in \mathbb{R}_+$,

$$\mathbb{E}\big[\exp\big(2c\, B_{\tau_a \wedge t} - 2c^2(\tau_a \wedge t)\big)\big] \;=\; 1.$$

(b) By letting $t \to +\infty$ and carefully justifying the passage to the limit, show that

$$\mathbb{P}(\tau_a < \infty) \;=\; e^{-2ca}.$$

Question 3 (law of the maximum).

(a) Justify the equality of events $\{S \geq a\} = \{\tau_a < \infty\}$, and deduce that S follows an exponential distribution whose parameter you will specify.

(b) Compute $\mathbb{E}[S]$ and $\mathbb{P}(S = 0)$.

Question 4 (memorylessness via the strong Markov property). We admit that the result of Question 2(b) remains valid for any barrier of level $b > 0$: $\mathbb{P}(\tau_b < \infty) = e^{-2cb}$, where $\tau_b = \inf\{t : X_t = b\}$. Let $a, b > 0$. Using the strong Markov property of Brownian motion applied to the stopping time τ_a, show *directly* (without using the exponential form of the law of S obtained in Question 3) that

$$\mathbb{P}\big(S \geq a + b \,\big|\, S \geq a\big) \;=\; \mathbb{P}(S \geq b),$$

and check that this result is consistent with Question 3.